Chapter 1: Large Numbers Around Us

1. **'X' is a 6-digit number formed using exactly three different digits. One of the digits appears once, another appears twice, and the third appears three times. When 2 lakhs are added to 'X', the resulting number 'Y' is still a 6-digit number. What is the highest possible value of 'Y'?**

   a) 977889 b) 999999 c) 999988 d) 999887

   Answer: c

   Solution:

   Given that X is a 6-digit number, where one digit occurs once, one digit occurs twice, and the other digit occurs thrice.

   Now, when 200000 is added to X, we get Y, which is also a 6-digit number.

   To maximise the value of Y, we have to maximise the value of X.

   Let X = __ __ __ __ __ __.

   The first blank cannot have 8 or more than 8, as adding 200000 + 800000 gives 1000000, which is the smallest 7-digit number. So, X must start with 7.

   X = 7 __ __ __ __ __.

   Also, we know that X is made of 3 different digits, where 1 digit occurs once, one digit occurs twice, and the other digit occurs thrice.

   Again, to maximise its value, the largest digit 9 must occur thrice, and the next largest digit 8 must occur twice.

   Hence, the HIGHEST possible value of X = 799988.

   Therefore, the HIGHEST possible value of Y is 799988 + 200000 = 999988.

   Option c is correct.

---

2. **What is the product of all the numbers on a telephone dial pad?**

   a) 12345 b) 32451 c) 362880 d) None of these

   Answer: d

   Solution:

   A telephone dial pad contains the numbers 0 to 9.
   When finding the product, all numbers are multiplied together.
   0 is included in the dial pad.
   We know that any natural number multiplied by 0 always results in 0.
   So, whatever the product of the other digits be, the total product of the numbers on the dial-pad becomes 0.
   As 0 is not present in the options, option d, 'None of these' is correct.

---

3. **Sam forms two different 6-digit numbers, X and Y, using digits from 0 to 9 (without repetition within a number). Exactly two digits are common between X and Y.**
   - **The digit 9 appears in the same position in both numbers**
   - **Y ends with 0, has only one even digit, and its digits are arranged in descending order**
   - **6 is present in X, and the number of even digits on its left is equal to the number of even digits on its right**

   If X is the largest possible number, what is the difference between X and Y?

   a) 9310 b) 9210 c) 8420 d) 8310

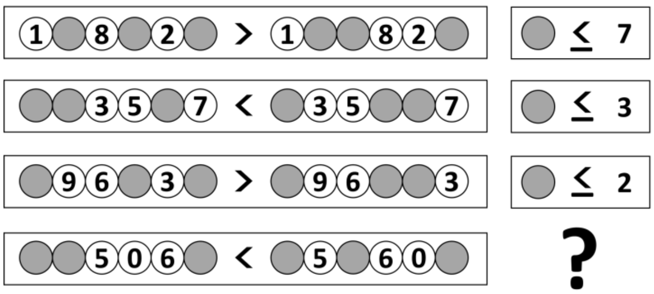
   Answer: a

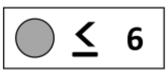
Solution:

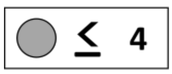
9 appears in the same position in both numbers.
Number Y ends with 0 and has only one even digit. So, the other digits are odd (as 0 is even) and arranged in descending order.
The odd digits in descending order are: 9, 7, 5, 3, 1.
So, **Y = 975310**.
In number X, the digit 9 will also be at the first position from the left (same as in Y).
So, **X = 9** _ _ _ _ _.
In number X, the digit 6 has an equal number of even digits on both sides, and X must be the largest possible number.
The even digits are: 0, 2, 4, 6, 8.
If 6 has only one even digit on each side, then number X will contain three even digits, and the remaining digits will be odd.
Since number Y contains all the odd digits and zero, three odd digits would be common between X and Y. However, only two common digits are allowed.
Therefore, 6 cannot have only one even digit on each side; it must have more than one.
If 6 has two even digits on both sides, then it must be at the hundreds position.
So, **X = 9 _ _ 6 _ _**.
The remaining even digits are: 8, 4, 2, 0. Arranging them to get the largest possible value of X, we get **984620**.
Now, find the difference:
984620 − 975310 = 9310
Therefore, the difference between the two numbers is **9310**. Hence, the correct answer is **option a**.

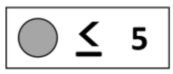
---

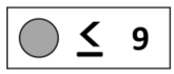
4. If each of the rows follows the same theme, what will come in place of "?"



a) 

b) 

c) 

d) 

Answer: c

Solution:

In each term, two six-digit numbers are formed by replacing the grey circle with a digit. The maximum possible value of the grey circle is given on the right. The comparison remains valid as long as the digit in the grey circle is less than or equal to the digit given on the right. For example, in the first term, the comparison holds when the grey circle is replaced with a digit less than or equal to 7.
178727 > 177827.
The condition is not valid for any number greater than 7.
In the question term, the numbers are: _ _ 506 _ < _ 5 _ 60 _.
Here, the comparison is valid as long as the grey circle is replaced with a digit less than or equal to 5.
555065 < 555605.
The condition is invalid for any other digit greater than 5. (For example, if the grey circle is replaced with 6, then 665066 < 656606 is invalid)
Hence, the correct answer is **option c**.

5. **What will come in place of "?"**

Crore - re	➡	00000
Million - ion	➡	000
Thousand - d	➡	00
Billion - llion	➡	0000
Lakh - kh	➡	?

a) 0 b) 00 c) 000 d) 0000

Answer: c

Solution:

Before the arrow, the **spelling of a number** is written, and a few letters from that spelling are shown after "-".

Rule:

- The **number of letters written after "-"** tells how many zeros are to be removed from that number.
- The **remaining zeros** are shown on the right side of the arrow.

Understanding the example

Crore = 1,00,00,000 : has **7 zeros**

Letters after "-" are **"re"**: 2 letters

Remaining zeros:

7 - 2 = 5: **00000**

Applying the rule to the question

Lakh = 1,00,000 : has **5 zeros**

Letters after "-" are **"kh"**: 2 letters

Remaining zeros:

5 - 2 = 3: **000**

Hence, option c is the correct answer.

---

6. **Six number tokens are given. Rearrange them to form a 6-digit number such that:**
   - **The difference between the first and last digits is as small as possible**
   - **The hundreds digit is double the thousands digit**
   - **No consecutive digits appear next to each other**
   - **All given tokens must be used exactly once**

   How many different 6-digit numbers between 2 lakhs and 8 lakhs can be formed under these conditions?

LEFT 2 5 8 1 4 7 RIGHT

Number Formed ___ ___ ___ ___ ___ ___

a) 1 b) 2 c) 3 d) More than 3

Answer: b

Solution:

As per the second condition, the hundreds digit is double the thousands digit.

So, there are only three possible cases, as per the available tokens.

Case A: __ __ 1 2 __ __

Case B: __ __ 2 4 __ __

Case C: __ __ 4 8 __ __

However, the third condition states that no consecutive digits of the number series appear next to each other. So, case A is invalid, as 1 and 2 (consecutive digits) are next to each other. Remaining cases:

Case B: __ __ 2 4 __ __

Case C: __ __ 4 8 __ __

Now, as per the first condition, the first and the last digits of the number have the least possible difference between them.

The least possible difference between any two tokens is 1.

From the above tokens, the pairs that have a difference of 1 are: (1, 2), (4, 5), and (7, 8).

As all tokens must appear once in a number, let's fill in cases B and C with the appropriate first and last digits, where the digits are not repeated.

Case B: As the middle digits of case B are 2 and 4, the only possible pair for the first and last digits is (7, 8)

a) 7 __ 2 4 __ 8
b) 8 __ 2 4 __ 7

Case C: As the middle digits of case C are 4 and 8, the only possible pair for the first and last digits is (1, 2)

a) 1 __ 4 8 __ 2
b) 2 __ 4 8 __ 1

Fill in the remaining blanks in such a way that no two consecutive digits of the number series appear next to each other.

Case B:

a) 7 5 2 4 1 8
b) 8 5 2 4 1 7

Case C:

a) 1 7 4 8 5 2
b) 2 7 4 8 5 1

Among all the numbers, the numbers that lie between 2 lakhs and 8 lakhs are 752418 and 274851.

Hence, option b is correct.

---

7. Two five-digit numbers are formed using different digits from 0 to 9. Both numbers have some digits missing, as shown below:

Number 1: _ 1 _ 9 _

Number 2: _ 4 _ 6 _

No two consecutive digits of the number series are present in the same number.
What is the maximum possible difference of the numbers formed?

a) 47525 b) 52663 c) 52429 d) 46733

Answer: b

Solution:

The digits not used in either number are 0, 2, 3, 5, 7, and 8.
No two consecutive digits are present in the same number.
Number 1 already contains 1 and 9. So, it can use 3, 5, and 7 only. (as 0, 2, and 8 are not allowed)
Number 2 already contains 4 and 6. So, it can use 0, 2, and 8 only. (as 3, 5, and 7 are not allowed)
To obtain the maximum difference, one number must be maximized and the other minimized.
We have to logically decide which number has a greater scope to be maximized and which can be minimized.

Clearly, we can see that the largest single digit '9' is already present in the tens place of number 1. So, the largest digit doesn't play any role in our current task.
Let's go to the next largest digit '8'.
We know that 8 belongs to number 2 and the largest place value of this number (the ten thousands) is also empty. So, placing 8 in the ten thousands place will significantly increase the overall value of number 2. (Hence, we get the maximum difference by having number 2 as large as possible and number 1 as small as possible).
Number 1: _ 1 _ 9 _
Number 2: 8 4 _ 6 _
Let's place the other digits (0 and 2) in number 2, accordingly.
Number 1: _ 1 _ 9 _
Number 2: 8 4 2 6 0
Similarly, to minimize number 1, we have to fill in the missing digits (3, 5, and 7) in ascending order.
Number 1: 3 1 5 9 7
Number 2: 8 4 2 6 0
The difference between both the numbers is: 84260 - 31597 = 52663.
Hence, the correct answer is option b.

---

8. **A, B, C, D, and E each have a different number card among one hundred, one thousand, ten thousand, one lakh, and ten lakhs.**
   - **C has an even number of zeros in his number**
   - **A's number has 2 zeros less than C's number**
   - **D's number has 4 zeros more than A's number**
   - **The number of zeros in B is more than that of A but less than C**

   Which number card does E have?

   a) One thousand b) Ten thousand c) One lakh d) Ten lakh

   Answer: c

   Solution:

   Since A has 2 fewer zeros than C, and D has 4 more zeros than A, C must lie between A and D, and must have an even number of zeros.

   Therefore, C has ten thousand, A has one hundred, and D has ten lakh.

   The number of zeros in B is more than that of A but less than C.

   B must fall between A and C. So, B has a thousand.

   The only card left for E is one lakh. So, the correct answer is option c.

---

9. **4 friends, Alex, Bob, Calvin and David have a different profession – Doctor, Lawyer, Cricketer, and Engineer and earn a different salary – ₹40,000, ₹50,000, ₹60,000 and ₹70,000 per year. (not necessarily in the same order)**
   - **The cricketer earns the highest**
   - **Alex earns more than Bob, while the doctor earns more than David, who is the engineer**
   - **Bob is the lawyer, and neither Bob nor David has an exact salary of ₹50,000**

   Which of these options shows the correct profession and salary of Calvin?

   a) Cricketer - ₹70,000 b) Doctor - ₹60,000 c) Doctor - ₹50,000 d) Doctor - ₹40,000

   Answer: c

   Solution:

   The 4 friends are Alex, Bob (lawyer), Calvin, and David (engineer). This leaves us with two professions, doctor and cricketer (for Calvin and Alex), which have not been fixed. It is mentioned that the Cricketer earns the highest.

Profession	Name	Salary
Cricketer		70,000
		60,000
		50,000
		40,000

Since the lawyer (Bob) and the engineer (David) do not earn ₹50,000 (as given in the statement, neither Bob nor David has an exact salary of ₹50,000), it must be the doctor who earns ₹50,000.

Profession	Name	Salary
Cricketer		70,000
		60,000
Doctor		50,000
		40,000

As "the doctor earns more than David, the engineer", David must earn ₹40,000, as the doctor earns ₹50,000. The only salary remaining for the lawyer is ₹60,000.

Profession	Name	Salary
Cricketer		70,000
Lawyer	Bob	60,000
Doctor		50,000
Engineer	David	40,000

Now, since Alex earns more than Bob, Alex must be the cricketer (₹70,000), and Calvin must be the doctor (₹50,000).

Therefore, Calvin's profession is a Doctor, with a ₹50,000 salary.

Profession	Name	Salary
Cricketer	Alex	70,000
Lawyer	Bob	60,000
Doctor	Calvin	50,000
Engineer	David	40,000

Option c is correct.

---

10. The following are the clues to the 6-digit password of Nisha's laptop:
35 tens
31 thousands
25 ones
423 hundreds
14 ten thousands
What is Nisha's password?

a) 213625　　b) 203675　　c) 212375　　d) 213675

Answer: d

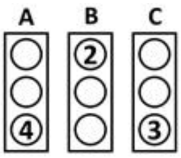
Solution:

To find Nisha's laptop password, add the values represented by each clue.

14 ten thousands = 14 × 10,000 = 1,40,000

31 thousands = 31 × 1,000 = 31,000
423 hundreds = 423 × 100 = 42,300
35 tens = 35 × 10 = 350
25 ones = 25
Now add all the values:
1,40,000 + 31,000 + 42,300 + 350 + 25 = 2,13,675
Therefore, Nisha's laptop password is 213675.
Hence, the correct answer is option d.

Columns A, B, and C contain 3 numbers each, such that:
- The number of times each number appears in all 3 columns combined, is equal to its value (e.g., 2 appears twice, 3 appears thrice, etc.)
- Each column contains AT LEAST TWO different numbers
If column A is the only column with THREE different numbers, then which column has the highest sum?



(a) Column A
(b) Column B
(c) Column C
(d) Both columns B and C

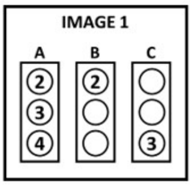
Answer: d

Solution:

From the question, we can say that **4 appears four times, 2 appears twice, and 3 appears thrice.**

It is stated that **Column A is the only column with three different numbers**, meaning Column A contains **2, 3, and 4**.

So, the empty circles of column A will have **2 and 3**.



We already have **two 2’s and two 3’s placed.**

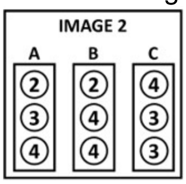
So, the remaining four circles must be filled with:

• One 3
• Three 4’s

Now, since Column B already contains a 2, we cannot place a 3 and 4 in it - doing so would make three different digits in Column B, which violates the condition that only Column A can have three different numbers.

Therefore, Column B must have **two 4’s.**

The remaining 3 and 4 will then go into Column C.



Sum of the digits of column A = 2 + 3 + 4 = 9
Sum of the digits of column B = 2 + 4 + 4 = 10
Sum of the digits of column C = 4 + 3 + 3 = 10

Hence, both columns B and C have the highest sum.

Hence, option d is the correct answer.